TIME SERIES ANALYSIS / LAB AAT

DAY-AHEAD ELECTRICITY PRICE FORECASTING

Volatility & prediction intervals with SARIMA–GARCH and quantile gradient boosting

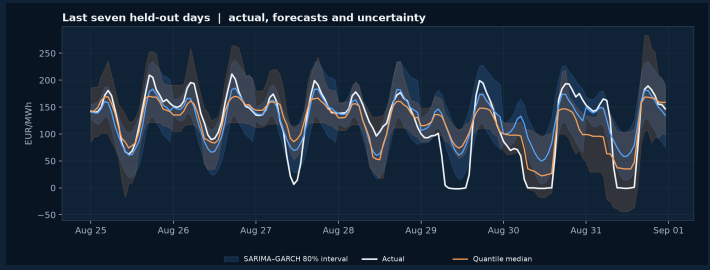
REAL DE-LU MARKET DATA

520-DAY CHRONOLOGICAL TEST

Presented by [Student name] • USN [USN]

B. M. S. College of Engineering | Department of Machine Learning

POINT FORECAST + UNCERTAINTY



A decision-support study — not a trading guarantee.

EXECUTIVE TAKEAWAY

One experiment. Two useful trade-offs.

A fair comparison on the same future-only test period.

BEST POINT ACCURACY

SARIMA–GARCH MAE
€23.28/MWh

CLOSEST TO 80% COVERAGE

Quantile boosting
79.8% observed coverage

LONG OUT-OF-SAMPLE TEST

520 days • 12,480 hourly forecasts

Bottom line **SARIMA–GARCH wins on point error; quantile boosting is slightly better calibrated and narrower.**

Evidence is limited to this zone, data window, feature set, and evaluation design.

Electricity prices are not an ordinary time series

The operational question

What might each hour cost tomorrow — and how uncertain is that estimate?

A point estimate alone hides the risk of being wrong.

APPLICATION: PROCUREMENT • FLEXIBLE DEMAND • STORAGE PLANNING

REPEATING DAILY SHAPE

Hourly cycles vary by time of day and day of week.

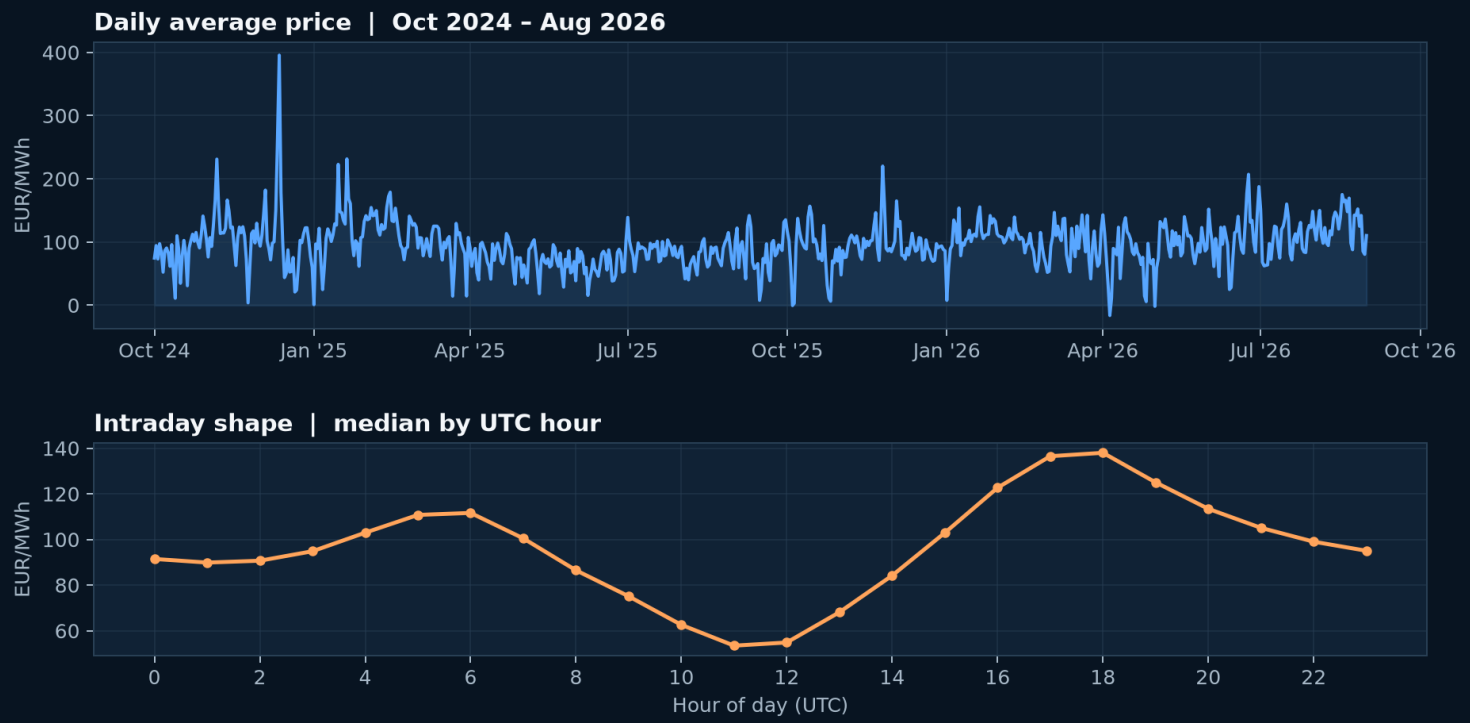
VOLATILITY CLUSTERS

Calm periods can be followed by sharp, irregular changes.

SPIKES & NEGATIVE PRICES

Outliers and prices below zero make naive averages risky.

A real, volatile market signal



MARKET

Germany–Luxembourg (DE-LU)
Energy-Charts / SMARD

OBSERVATIONS

16,800 hourly prices
1 Oct 2024 – 31 Aug 2026 (UTC)

PREPARATION

15-minute values averaged hourly; negative prices retained.

Source: Energy-Charts price API • Attribution: Bundesnetzagentur / SMARD • CC BY 4.0

LITERATURE & RESEARCH GAP

What prior work tells us — and what this study tests

PRICE DEPENDENCE

Autoregressive structures capture useful intraday effects. [1, 2]

VOLATILITY MATTERS

GARCH is a recognized tool for changing price uncertainty. [3]

PROBABILISTIC VALUE

Evaluate both calibration and interval sharpness—not error alone. [4, 5]

CLASSROOM-SCALE GAP

Put a seasonal SARIMA–GARCH interval and a direct quantile interval through the same long, chronological DE-LU test.

This is a focused comparison, not a claim that the model combination is new or universally superior.

LITERATURE: GARCIA ET AL. (2005); ZIEL (2016); NOWOTARSKI & WERON (2018); UNIEJEWSKI ET AL. (2018); LAGO ET AL. (2021)

EVALUATION DESIGN

Test the future as it would arrive

700 complete UTC days • No random shuffle

INITIAL TRAINING

180 days

ROLLING HOLDOUT • 520 DAYS • 12,480 HOURLY FORECASTS

30 Mar 2025 → 31 Aug 2026

ORIGIN

At each day boundary, use information available before delivery.

FORECAST

Predict the next 24 hourly prices and interval bounds.

UPDATE

Only after forecasting, add the day's actuals to history.

Leakage control: day-ahead target values never build that same day's input features.

Baseline: model the mean, then its changing variance

SEASONAL MEAN

SARIMA(1,0,1) × (1,0,1,24) models dependence and the daily cycle.

CONDITIONAL VOLATILITY

GARCH(1,1) updates uncertainty using recent forecast residuals.

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$$

80% interval ≈ forecast mean ± 1.645 × forecast standard deviation

WHY THIS BASELINE?

Interpretable statistical model with explicit daily seasonality.

GARCH widens or narrows bounds as estimated residual volatility changes.

Normal standardized innovations are an assumption; test coverage checks calibration.

Enhancement: learn the price distribution directly

Three quantile learners

τ = 0.10	τ = 0.50	τ = 0.90
LOWER BOUND	MEDIAN	UPPER BOUND

10th–90th percentile = nominal 80% prediction interval

PAST-PRICE SIGNALS

24h, 48h and 168h lags • trailing mean and standard deviation

CALENDAR SIGNALS

Hour, weekday and month cycles • 150 shallow boosting trees

Quantile loss allows asymmetric bounds; it does not assume errors are normally distributed.

REPRODUCIBLE PROTOCOL

Same dates. Same targets. Same scorecard.

01

OBSERVE

Public hourly DE-LU prices
UTC, 2024–2026

02

TRAIN

180 initial days chronological history

03

FORECAST

Two models × 24 hours for each test day

04

UPDATE

Add actuals after the forecast is complete

05

EVALUATE

Point error + interval coverage, width, loss

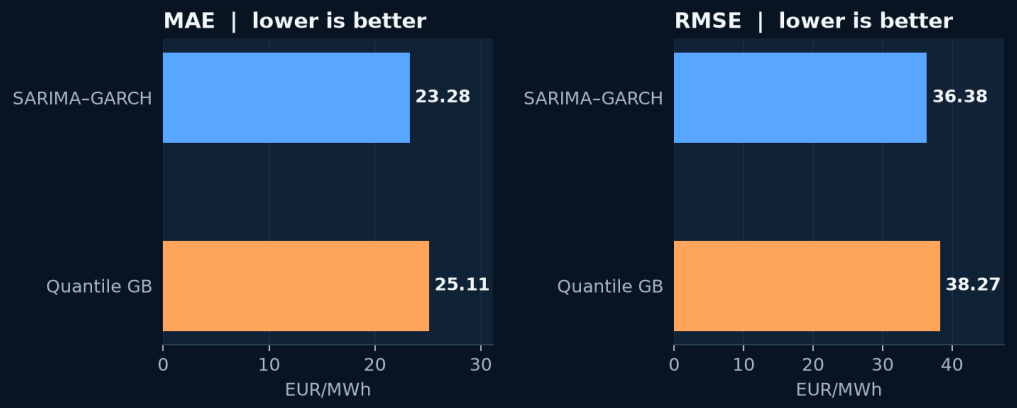
POINT FORECAST METRICS

MAE • RMSE • sMAPE (instead of MAPE for zero/negative prices)

INTERVAL METRICS

80% coverage • mean width • 10th / 90th pinball loss

Results: accuracy vs. interval calibration



TEST METRIC	SARIMA–GARCH	QUANTILE GB
80% interval coverage	82.9%	79.8%
Mean interval width	€83.25/MWh	€81.90/MWh
10th-percentile pinball	6.43	6.77
90th-percentile pinball	6.30	6.09

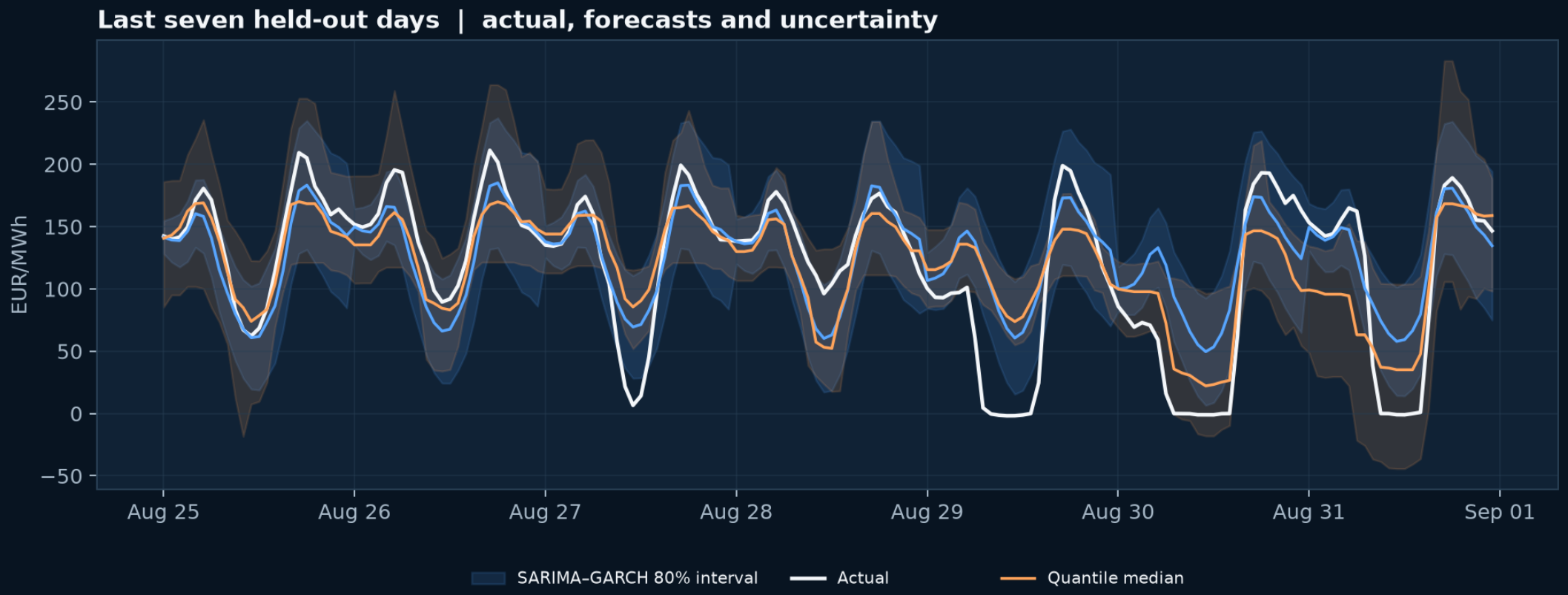
Point error favors SARIMA–GARCH (23.28 vs. 25.11 MAE).

Quantile coverage: 79.8% (nominal target 80%).

Coverage and width must be read together: neither metric alone proves an interval is useful.

ACTUAL VS. FORECAST · LAST 7 TEST DAYS

What the forecasts look like in practice



BLACK ACTUAL

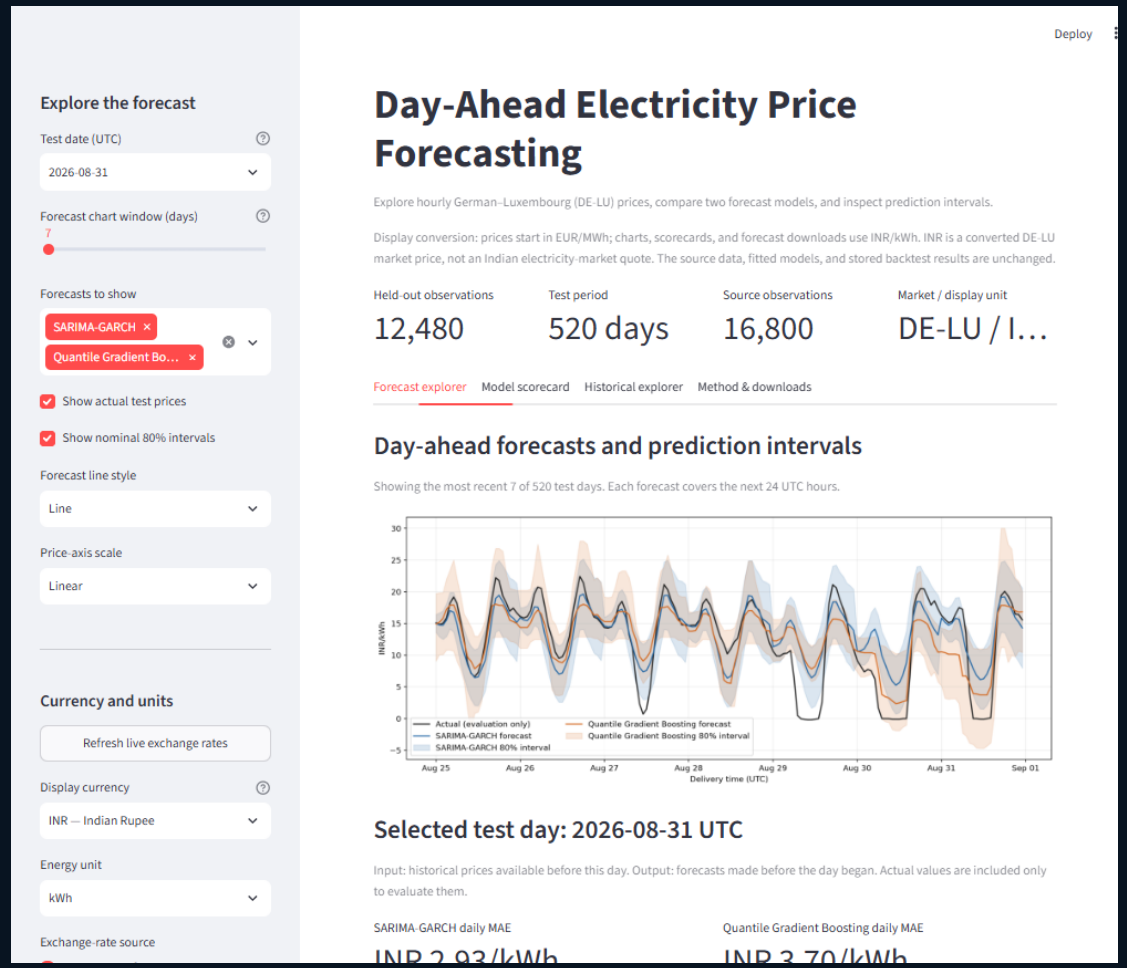
BLUE SARIMA–GARCH

ORANGE QUANTILE MEDIAN

Shaded bands = nominal 80% prediction intervals. Spikes remain hard to anticipate.

INTERACTIVE DASHBOARD

A working input → output demo



INPUT

Choose a held-out UTC test date.

OUTPUT

24 hourly forecasts + 80% bounds.

EVALUATE

Actuals and daily errors shown separately.

INR is a display conversion of DE-LU prices—not an Indian market quote.

Live demo: select 2026-08-31 UTC → inspect the hourly table → explain actuals are for evaluation only.

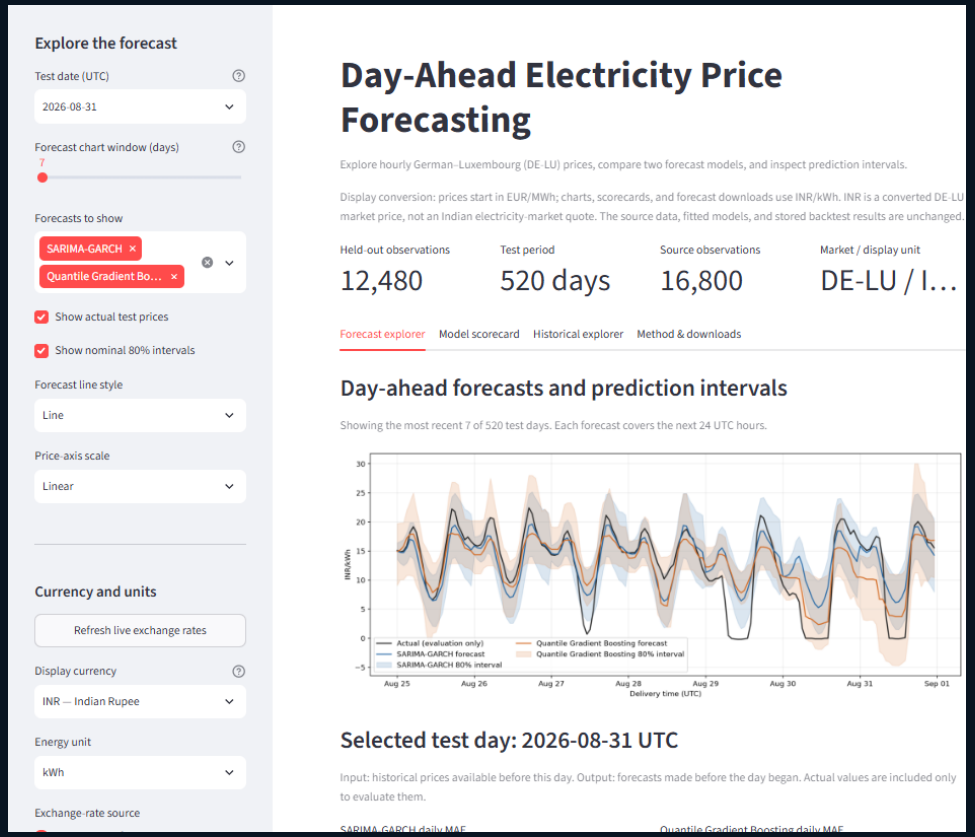
DASHBOARD SCREENSHOTS

Inspect a test day, hour by hour

SELECTED DATE + FORECAST TABLE

Timestamp (UTC)	Actual (INR/kWh)	SARIMA-GARCH forecast	SARIMA-GARCH lower 80%	SARIMA-GARCH upper 80%	Quantile Gradient Boosting forecast	Quantile Gradient Boosting low
2026-08-31 00:00:00+00:00	16.256	15.864	13.852	17.876	10.518	
2026-08-31 01:00:00+00:00	15.679	15.186	12.817	17.555	10.395	
2026-08-31 02:00:00+00:00	15.107	14.748	12.069	17.427	10.157	
2026-08-31 03:00:00+00:00	15.298	15.049	12.092	18.006	10.157	
2026-08-31 04:00:00+00:00	16.47	15.833	12.623	19.044	10.157	
2026-08-31 05:00:00+00:00	17.5	15.659	12.213	19.105	10.023	
2026-08-31 06:00:00+00:00	17.225	13.345	9.679	17.011	6.702	
2026-08-31 07:00:00+00:00	13.423	10.655	6.781	14.528	6.702	
2026-08-31 08:00:00+00:00	4.068	9.3	5.23	13.371	5.569	
2026-08-31 09:00:00+00:00	0.002	7.888	3.629	12.146	3.949	
2026-08-31 10:00:00+00:00	-0.006	6.799	2.36	11.237	3.877	
2026-08-31 11:00:00+00:00	-0.1	6.147	1.536	10.758	3.726	
2026-08-31 12:00:00+00:00	-0.109	6.286	1.508	11.064	3.726	
2026-08-31 13:00:00+00:00	-0.009	7.047	2.108	11.987	3.726	
2026-08-31 14:00:00+00:00	0.106	8.428	3.332	13.523	5.059	
2026-08-31 15:00:00+00:00	7.746	12.611	7.364	17.857	10.457	
2026-08-31 16:00:00+00:00	16.754	17.096	11.702	22.489	16.648	
2026-08-31 17:00:00+00:00	19.425	19.136	13.599	24.673	17.864	
2026-08-31 18:00:00+00:00	20.066	19.192	13.515	24.868	17.864	
2026-08-31 19:00:00+00:00	19.338	18.142	12.329	23.955	17.706	
2026-08-31 20:00:00+00:00	18.199	17.155	11.209	23.101	17.573	
2026-08-31 21:00:00+00:00	16.473	15.888	9.812	21.964	16.989	
2026-08-31 22:00:00+00:00	16.399	15.202	8.998	21.405	16.806	
2026-08-31 23:00:00+00:00	15.553	14.282	7.953	20.61	16.857	

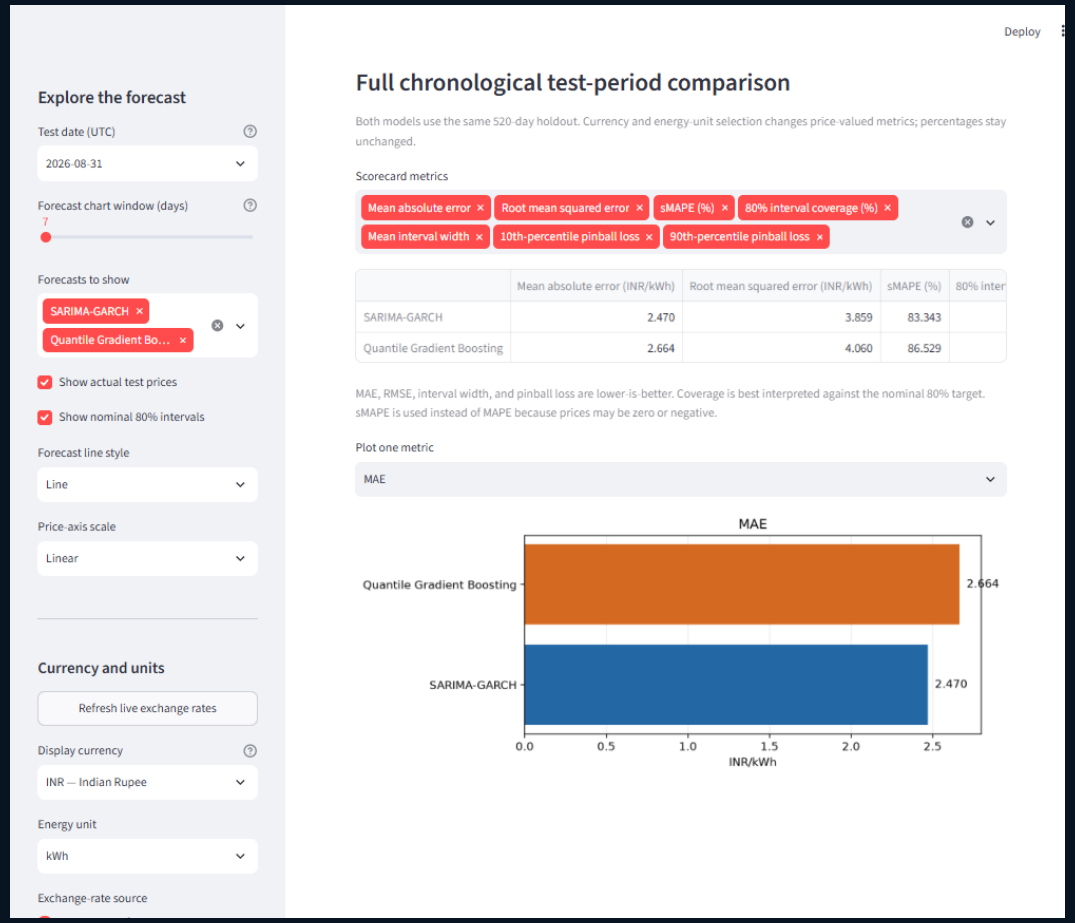
ACTUALS VS. FORECASTS + INTERVALS



Actuals are for evaluation only. Scroll the live table to inspect all 24 hourly rows.

DASHBOARD SCREENSHOT

Compare models in the live scorecard



LOWER POINT ERROR

SARIMA–GARCH MAE: €23.28/MWh

COVERAGE NEAR 80%

Quantile boosting: 79.8%

Select a metric in the app to compare scorecards in the chosen display currency.

LIMITATIONS & NEXT STEPS

Clear limits make the result more credible

SINGLE ZONE + HISTORICAL WINDOW

One DE-LU holdout cannot establish general performance.

PRICE-ONLY FEATURES

No weather, demand, generation, fuel or neighboring-zone inputs.

UTC DELIVERY DAYS

UTC 24-hour blocks differ from local market days at daylight-saving changes.

INTERVAL ASSUMPTION

Normal standardized GARCH innovations may miss asymmetric tails.

NEXT **Add external drivers • test multiple zones/seasons • calibrate intervals • evaluate decision value**

SDG 7 is the primary connection; SDG 13 is secondary through energy-system planning. Forecasts alone do not achieve either goal.

CONCLUSION

Takeaway: forecast the price and the risk

01 **Chronological testing makes the comparison meaningful.**

02 **SARIMA–GARCH delivered lower point error in this holdout.**

03 **Quantile boosting gave coverage closer to the 80% target.**

The right model depends on the decision and the cost of forecast errors.

QUESTIONS?

Thank you • [Student name] • [USN]

SOURCES

References & data attribution

[1] Garcia et al. (2005). A GARCH forecasting model to predict day-ahead electricity prices. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2005.846044

[2] Ziel (2016). Forecasting electricity spot prices using Lasso. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2016.2521545

[3] Nowotarski & Weron (2018). Recent advances in electricity price forecasting: probabilistic forecasting. Renewable & Sustainable Energy Reviews. doi:10.1016/j.rser.2017.05.234

[4] Uniejewski, Weron & Ziel (2018). Variance stabilizing transformations for electricity spot price forecasting. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2017.2734563

[5] Lago et al. (2021). Forecasting day-ahead electricity prices: algorithms, best practices and open-access benchmark. Applied Energy. doi:10.1016/j.apenergy.2021.116983

Data: Energy-Charts price API, DE-LU bidding zone; attribution to Bundesnetzagentur / SMARD; CC BY 4.0. https://api.energy-charts.info/

Methods/software: statsmodels SARIMAX; arch GARCH; scikit-learn GradientBoostingRegressor quantile loss.

All performance values are computed from the included 520-day held-out predictions.